



PROFESSIONAL PORTFOLIO

Ten Years Building the Systems Enterprises Run On

IT asset & configuration management, Power Platform automation, and AI-assisted engineering — from warehouse floor to leading enterprise account delivery, solo-built and production-proven.

Madhankumar · IT Operations, Automation & Power Platform Specialist

10+

YEARS IN ENTERPRISE IT
OPERATIONS

9

PRODUCTION SYSTEMS BUILT SOLO

0.5 → 0.1

FTE REPORTING EFFORT, ONE
AUTOMATION

Client, employer, and colleague names generalized throughout this document as a standard confidentiality practice. Every figure, timeline, and technical detail below is real.



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01 · CAREER JOURNEY

A decade, one direction

I started on a warehouse floor managing physical stock. Ten years later, I personally own IT asset and configuration governance for six enterprise client accounts, lead a team of analysts, and build the automation that replaces the manual work I used to do by hand. **Every system in this portfolio was built to solve a problem I'd lived inside myself.**

2015

Warehouse & Stock Operations

Managed physical stock receipt, dispatch, and returns; coordinated parts allocation with field engineers and logistics teams.

2015 – 2017

IT Service Delivery & Supply Chain Operations

Processed 100+ monthly IT orders, mobility requests, and PO approvals; owned asset data validation and reconciliation. Promoted for performance.

2017 – 2019

Global Network Procurement & Mobility Operations

Managed network contracts and telecom order lifecycles for enterprise international accounts; bulk CMDB reconciliation at scale.

2020

IT Asset Management — Individual Contributor

Moved into dedicated CMDB validation and asset governance work. Performance here moved me onto a leadership track.

2021 – PRESENT

Team Lead & Enterprise Account Owner

Leading a small analyst team while personally owning CMDB validation, audit reporting, and stakeholder delivery for six enterprise client accounts. Ten years of service reached in 2025.

6

ENTERPRISE ACCOUNTS
PERSONALLY OWNED

10+ yrs

CONTINUOUS IT OPERATIONS
EXPERIENCE

0

ESCALATIONS ON OWNED
ACCOUNTS, 2025



02 · CORE COMPETENCIES

Five domains, one working style

I build things that keep working after I leave the room — documented, automated, and paired with the reporting layer that makes them useful to a team, not just to me.



IT Asset & CMDB Management

Ten-plus years of hands-on CMDB validation, asset lifecycle governance, and audit reporting across enterprise client portfolios.

ServiceNow CMDB

Asset Lifecycle

Data Reconciliation

Audit Reporting



Power Platform Development

Canvas Apps, Power Automate flows, and Power BI reporting built solo — from cascading forms to full data-entry-to-dashboard pipelines.

Power Apps

Power Automate

Power BI

SharePoint



Python Automation & Reporting

Flask and Streamlit applications automating multi-source data extraction, validation, and report generation for production use.

Python / Flask

Streamlit

openpyxl

Self-Healing Deployment



Team Leadership & Mentoring

Day-to-day leadership of an analyst team — work allocation, coaching, escalation handling, and onboarding training.

Work Allocation

Coaching

Knowledge Transfer

Onboarding



AI-Assisted Development

I use AI as a real development accelerator — not a buzzword. Most of my Python tools were built with AI-assisted coding; one runs a live LLM at runtime. I'm precise about the difference, and I'd rather under-claim than oversell it.

AI-Assisted Coding

Fuzzy Matching & Confidence Scoring

LLM API Integration

Prompt Engineering



03 · FLAGSHIP PROJECT

An Automated ITAM Reporting Platform

What started as one client's manual reporting fix became a production platform used by three separate teams. This is the project I'm proudest of — and the clearest proof of how I build.

PHASE 1 · MAR–MAY 2026

Sole builder
Python / Flask

Automated Multi-Report ITAM Pipeline

One enterprise account's monthly ITAM reporting was a fully manual, multi-step process — pulling from a jump server, staging on a local PC, assembling eight separate report categories by hand, then uploading to SharePoint. I rebuilt the entire chain as an automated pipeline.

WHAT IT DOES

- Extracts from four separate source systems automatically
- Validates data — duplicate detection, retired-asset checks, CPU/spec checks
- Auto-generates all 8 report categories and delivers directly to SharePoint
- Full audit log, with a self-healing deployment layer

HARDEST PROBLEM

- Not one hard problem but several stacked together: multi-source extraction and validation logic, self-healing deployment, and exact-format report output all had to work together reliably, unattended, every month

~48 hrs

MEASURED SAVING / YEAR ON
GENERATION ALONE

0.5 → 0.1

FTE, PROJECTED FULL REPORTING
EFFORT

8

REPORT CATEGORIES AUTOMATED

Python / Flask

Multi-Source Extraction

Data Validation Engine

SharePoint Delivery

Audit Logging



03 · FLAGSHIP PROJECT (CONTINUED)

From One Account to a Cross-Team Platform

PHASE 2 · APR–JUL 2026

Sole builder
Flask / Python / openpyxl

Generalized Into a Reusable Platform

Once the first automation proved out, I rebuilt it as a general-purpose platform — still covering all 8 report types, but now deployable by non-technical users on their own machines, with automatic setup and dependency detection.

WHAT CHANGED

- Self-healing local Windows deployment — automatic Python/package detection and setup for users with no technical background
- One-click installer wizard replacing manual environment setup

ADOPTION & IMPACT

- 11 total users across my own team, the client-side team, and other regional teams
- Reporting cycle: **4 hours** → **10 minutes**
- Actively used daily/weekly in production today

Self-Healing Installer

Non-Technical User Deployment

Cross-Team Adoption

Why this is the flagship: it's not a demo — it's a real automation that started by solving my own account's problem, then proved valuable enough that three separate teams adopted it independently.



04 · POWER PLATFORM

Built Under Real Deadlines, Still in Production

CANVAS APP · EXCEL ONLINE

Jun–Jul 2026
4-day turnaround

Client Asset Forecast Form — Built in 4 Days

A direct request from my manager: a client team was doing manual SharePoint data entry and needed an interactive solution — fast. Call received at 3:39 PM; a fully interactive Power Apps solution was in production within four working days.

WHAT IT DOES

- Cascading dropdown selection with a multi-select gallery interface
- Excel Online as the live data source, with full submission validation logic

OUTCOME

- Live in production, actively used by the client team
- Recognized internally with a company delivery award for speed and quality

Power Apps Canvas

Excel Online

Cascading Dropdowns

CANVAS APP · POWER AUTOMATE · POWER BI

32-day delivery
Sept – Oct 2025

Multi-Region Monthly Asset Tracking System

A monthly asset-tracking Canvas App used by analysts covering 146 regional client sites across Asia-Pacific — structured entry feeding a dual Power BI dashboard and two automated stakeholder report emails, every month.

WHAT IT DOES

- 10 fields + 3 conditional sub-fields, fixed monthly deadline built into the workflow
- SharePoint list architecture optimized to hold up at 146-client scale

OUTCOME

- 100% monthly compliance — all 146 clients reporting consistently, every month
- Two automated report emails to regional heads and managers, zero manual consolidation

Power Apps Canvas

SharePoint at Scale

Power BI Dual Dashboards

Automated Email Routing



04 · POWER PLATFORM (CONTINUED)

Two more systems, one legacy foundation

API INTEGRATION · BULK VALIDATION

Bulk Asset Intake & Validation Pipeline

Dec 2025 – Jan 2026

A client-facing Power Apps intake form feeding a bulk validation and load pipeline for configuration-item requests, converting validated records into a fixed XML template for bulk loading.

- Sole builder of the Power Apps intake form; API integration to the downstream asset database built jointly with another engineer
- Hardest problem: bulk-load performance at scale

SHAREPOINT · 2017 · LEGACY

Self-Service Disconnect Request Portal

Mar – May 2017

No self-service channel existed for global users to request disconnection of telecom lines — every request came in directly, creating a constant bottleneck.

- Built on a brand-new platform in 2017, with no AI tools or technical guidance available — self-taught under pressure
- Personally the most technically challenging project of my career



05 · AI-ASSISTED ENGINEERING

Where AI actually runs — and where it doesn't

I'd rather be precise than impressive. Most of my automation uses AI-assisted *development*; only one project below runs live AI *inference*. Here's the honest breakdown.

PYTHON · STREAMLIT · FUZZY MATCHING

Oct – Dec 2025

Asset Data Validation Tool

Built to close a real gap in asset data validation — a Streamlit application using fuzzy matching, confidence scoring, and multi-engine search to validate asset records. Built with AI-assisted coding; runtime logic is deterministic (fuzzy-match algorithms and confidence scoring, not live LLM calls).

RESULT

- 95% internal validation accuracy
- Live ServiceNow API integration attempted — blocked by internal approval, not by capability

ADOPTION

- Used initially by my own team
- Now actively used by a separate regional team in another country

Python

Streamlit

Fuzzy Matching

Confidence Scoring

PYTHON · LIVE LLM API

Personal build

Personal AI Assistant — Genie

A Python chatbot built solo, calling a live LLM API at runtime — the one project in this portfolio where AI inference actually happens live, not just at build time.

STATUS

- Active, used daily/regularly for personal productivity

WHY IT'S HERE

- Demonstrates real, hands-on LLM API integration — not just AI-assisted coding of other tools

Python

LLM API Integration

Personal Productivity Tool

Personal-use tool, not a client deliverable — included to show real hands-on capability with live AI inference, distinct from the AI-assisted (but deterministic) automation elsewhere in this portfolio.



06 · INDEPENDENT BUILDS

Proof I can ship, start to finish

Outside client and employer work, I build to learn — and to prove I can take something all the way to a finished, working product.

KOTLIN · JETPACK COMPOSE · ROOM DB

Personal project · v1.0 shipped

ChicoCard — Offline Learning App

A fully offline Android flashcard/learning app, built solo from a personal learning need — and taken all the way to a tagged, finished v1.0 release, not left as a prototype.

WHAT IT DOES

- Fully offline-first architecture — no network dependency
- State management via StateFlow, local persistence via Room DB

HARDEST PROBLEM

- Offline-first architecture and state management, done right, without a backend to lean on

Kotlin

Jetpack Compose

Room DB

StateFlow

Offline-First

Why it's here: most side projects stay half-finished. This one didn't — it's proof I can take a build all the way to a real, tagged release on my own.



07 · DOMAIN EXPERTISE

IT Asset & Configuration Management

The operational depth behind every automation in this portfolio — ten-plus years of hands-on ownership of asset accuracy, configuration data, and enterprise client reporting.

6

ENTERPRISE ACCOUNTS PERSONALLY
OWNED

95%+

ASSET ACCURACY MAINTAINED IN
CMDB

100%

SLA COMPLIANCE, 2025, OWNED
ACCOUNTS

Where I've operated

- CMDB validation, tracking, and lifecycle management across six enterprise client accounts
- Client audit reporting and stakeholder-facing governance documentation
- Bulk data validation and structured-load pipelines for configuration-item records
- Integration between asset intake tools and enterprise ITSM platforms

Measured account outcomes

- Asset data completeness improved from 60% to 90% on one managed account
- Configuration item data quality improved from 90% to 100%
- Hardware end-of-life compliance improved from 90% to 100%
- Manual reporting effort cut by up to 70% (up to 30 hrs/month) across managed accounts

Why this matters for automation work: I'm not learning the business problem while I build. I've lived inside asset and configuration management operations for a decade — when a client describes a messy asset-tracking or reporting process, I recognize the shape of it immediately.



08 · LEADERSHIP & TEAM IMPACT

Two scopes, both mine

Day-to-day team leadership and personal enterprise-account ownership are separate responsibilities — I hold both, and I'm precise about where each one ends.

Team Leadership

- Day-to-day leadership of an analyst team (currently 3 members)
- Work allocation, informal coaching, and escalation handling
- Formal onboarding training for new joiners on core tools
- Formal performance ratings sit with my manager — my scope is allocation and coaching, not HR rating authority

Individual Account Ownership

- Personal, direct ownership of six enterprise client accounts — separate from the team's shared workload
- CMDB validation, audit reporting, and stakeholder delivery, end to end
- Zero escalations, 100% SLA compliance across owned accounts in 2025

KNOWLEDGE TRANSFER & MENTORING

- Organized and delivered full training and handover for a major account transition
- Mentored team members through account transitions across three separate client accounts
- Ongoing stakeholder guidance and reporting-process training for new team members

"His guidance and support made a difference" — colleague feedback on day-to-day mentoring and support.



09 · RECOGNITION, CERTIFICATIONS & EDUCATION

Formally recognized, formally trained

2

FORMAL WORKPLACE RECOGNITION
AWARDS

10 yrs

SERVICE MILESTONE REACHED, 2025

6

CERTIFICATIONS & FORMAL
TRAINING COMPLETED

Recognition

- Company delivery recognition award — for a 4-day Power Apps build, nominated by my manager
- Security awareness recognition — correctly identified all 6 consecutive simulated phishing tests, FY25
- 10 Years of Service milestone, reached September 2025

Certifications & Training

- ITIL 4 Fundamentals — Essentials of IT Service Management (2024)
- Certified Scrum Product Owner (CSPO) — in progress
- Generative AI & Prompt Engineering — completed
- Power BI Bootcamp & Machine Learning Bootcamp — completed

EDUCATION

B.C.A.

BACHELOR OF COMPUTER APPLICATIONS — PERIYAR
UNIVERSITY · FIRST CLASS WITH A+ GRADE · 2011

M.C.A.

MASTER OF COMPUTER APPLICATIONS — ANNA
UNIVERSITY, CHENNAI · 2011–2014



10 · LET'S TALK

What ties this portfolio together

Every system in this portfolio was built solo, deployed into a real production environment with real users and real deadlines — and grounded in ten years of actually working inside the problems I now automate. I don't just design a workflow; I've lived it, then built the thing that replaces it.

IT Asset & CMDB Management

Power Apps (Canvas)

Power Automate

Power BI

SharePoint

Python (Flask, Streamlit)

Excel VBA & Power Query

AI-Assisted Development

LLM API Integration

Team Leadership

Stakeholder Reporting

Have a process that shouldn't be manual anymore?

Whether it's asset & configuration tracking, a Power Apps workflow, or an Excel/Python reporting pipeline — I'd be glad to look at your specific requirements and tell you honestly how I'd approach it.